

Wiener-to-Hilbert–Schmidt Transfer for Multiplicative Resonance: Width-Sensitive Dispersion and the Long-Fiber Gate

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Abstract

TPC-99 diagonalizes the multiplicative resonance carrier but bounds its nonprincipal part by the largest incomplete-character eigenvalue, costing $\sqrt{q} \log q$ at every resonance width. That estimate treats the low-window reconstruction weight as an arbitrary L^2 vector. In the literal problem this weight is fixed and has bounded Wiener mass. We exploit that asymmetry.

For $G = \mathbb{F}_q^\times$, $E_H = \{\pm 1, \dots, \pm H\}$, and

$$\mathcal{I}_H(a, b) = \sum_{\omega, r \in G} a(\omega) b(r) \mathbf{1}_{E_H}(r\omega),$$

write $\mathcal{R}_H(a, b)$ for the nonprincipal part. We prove that the exact norm of this functional in the centered a variable is

$$\mathfrak{S}_H(b)^2 = \frac{1}{q-1} \sum_{\chi \neq \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 |\widehat{b}(\overline{\chi})|^2.$$

Consequently

$$|\mathcal{R}_H(a, b)| \leq \|b\|_1 \sqrt{\frac{2H(q-1-2H)}{q-1}} \|a^\circ\|_2.$$

The factor is exact using only $\|b\|_1$: a point mass b attains it. It vanishes at $H = (q-1)/2$, and for the literal width $H_q(N)$ it is at most $\sqrt{2q/N}$ whenever $N \geq 3$.

Applied to the corrected frequency-free TPC-99 census and its fixed filter $b_X = |\mathfrak{m}_{K,X}|$, this replaces the uniform $Q_X^{3/2}$ centered-energy target by the weaker width-sensitive gate

$$\sum_H \sqrt{\frac{2H(q_X-1-2H)}{q_X-1}} \|a_H^\circ\|_2 \ll X^{o(1)} Q_X^2.$$

After TPC-100 separately returns its $q_X \mid u$ subcarrier, its exact singleton-cell decomposition produces separate width-weighted atomic-diagonal and cross-cell gates. The singleton fact itself supplies no dispersion. We also prove an abstract principal-to-diagonal transfer under a stated maximum-atom bound. The finite-group theorem is L0, and the literal crosswalk is L1. The principal mass, literal atom cap, and cross-cell estimate are not proved here; no L2 fixed-shift cancellation or prime-pair conclusion is claimed.

Keywords: multiplicative resonance; Hilbert–Schmidt spectrum; Wiener norm; affine Möbius fiber; width-sensitive dispersion; no-wrap prime.

Literal-frame convention. One prescribed nonzero h_0 , both polarizations, every frequency-free branch key, the actual support and masks, exact content, and one global normalization are retained. The finite group theorem below is returned to the literal positive carrier only through an explicit crosswalk.

1 Introduction

The literal low-window export retains the original fixed nonzero shift, source rows, exact-content branches, projector children, actual masks, and reconstruction multiplier [2, 3]. After the conductor-one branch is returned, TPC-99 identifies every remaining resonant nonconstant branch with an incidence condition

$$r\omega_\beta \in E_H \subset \mathbb{F}_{q_X}^\times.$$

Its multiplicative-character theorem is exact. The quantitative return used there, however, invokes the largest nonprincipal singular value

$$\max_{\chi \neq \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)| \ll \sqrt{q} \log q$$

and then pays this worst value against the whole centered multiplier census [4].

There is an unused asymmetry. The multiplier census a_H is the unknown growing object, but the frequency weight

$$b_X(r) = |\mathbf{m}_{K,X}(r)| \mathbf{1}_{\{0 < d_{q_X}(r,0) \leq R_X\}}$$

is fixed by the zero-arc reconstruction and satisfies $\|b_X\|_1 \ll_K 1$. Instead of bounding the operator uniformly for all b , one may compute the exact norm of the scalar functional $a \mapsto \mathcal{R}_H(a, b_X)$. Cauchy–Schwarz then falls on the *whole* nonprincipal spectrum, whose squared mass is known exactly:

$$\sum_{\chi \neq \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 = 2H(q-1-2H).$$

This exchanges a worst-eigenvalue cost for a Wiener-to-Hilbert–Schmidt transfer.

1.1 What changes

Define

$$g_q(H) = \sqrt{\frac{2H(q-1-2H)}{q-1}}.$$

The old nonprincipal return was

$$|\mathcal{R}_H(a_H, b_X)| \ll_K \sqrt{q} \log q \|a_H^\circ\|_2.$$

The new one is

$$|\mathcal{R}_H(a_H, b_X)| \ll_K g_q(H) \|a_H^\circ\|_2.$$

This is never worse up to harmless constants and is asymptotically better when $H = o(q)$; the improvement is polynomial when $H \leq q^{1-\delta}$ for a fixed $\delta > 0$, apart from logarithms. It also records the exact endpoint: $g_q((q-1)/2) = 0$, because $E_H = G$ and the incidence operator is purely principal.

1.2 What does not change

The principal term

$$\frac{2H}{q-1} \|a_H\|_1 \|b_X\|_1$$

is untouched. Nor does the theorem estimate the native census. TPC-100 decomposes its centered energy into an opened-atom diagonal ledger and a cross-cell collision excess [1]; the present theorem merely assigns the correct width cost to those quantities. Thus it opens a wider analytic door without relabeling an interface result as fixed-shift arithmetic cancellation.

1.3 Organization

Section 2 proves the exact functional norm and its Wiener bound. Section 3 proves sharpness and endpoint behavior. Section 4 returns the estimate to the frequency-free literal census. Section 5 records the resolved-length gain, and section 6 combines it with the TPC-100 cell ledger. The remaining sections give the finite certificate, route ledger, and claim boundary.

2 The exact fixed-filter functional norm

Let q be an odd prime, put $G = \mathbb{F}_q^\times$, and write $n = q - 1$. For $1 \leq H \leq n/2$, set

$$E_H = \{\pm 1, \dots, \pm H\} \subset G.$$

For $f : G \rightarrow \mathbb{C}$, use

$$\widehat{f}(\chi) = \sum_{x \in G} f(x) \overline{\chi(x)}, \quad f^\circ = f - \frac{1}{n} \left(\sum_x f(x) \right) \mathbf{1}_G.$$

Parseval reads

$$\sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 = n \sum_{x \in G} |f(x)|^2.$$

TPC-99 proves the exact decomposition

$$\mathcal{I}_H(a, b) = \frac{2H}{n} \left(\sum_{\omega} a(\omega) \right) \left(\sum_r b(r) \right) + \mathcal{R}_H(a, b), \quad (1)$$

$$\mathcal{R}_H(a, b) = \frac{1}{n} \sum_{\chi \neq \chi_0} \widehat{\mathbf{1}_{E_H}}(\chi) \widehat{a}(\overline{\chi}) \widehat{b}(\overline{\chi}). \quad (2)$$

Definition 2.1 (Fixed-filter spectral mass). For $b : G \rightarrow \mathbb{C}$, define

$$\mathfrak{S}_H(b)^2 = \frac{1}{n} \sum_{\chi \neq \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 |\widehat{b}(\overline{\chi})|^2. \quad (3)$$

Theorem 2.2 (Exact fixed-filter norm). For every $a, b : G \rightarrow \mathbb{C}$,

$$|\mathcal{R}_H(a, b)| \leq \mathfrak{S}_H(b) \|a^\circ\|_2. \quad (4)$$

For fixed b , the constant $\mathfrak{S}_H(b)$ is the exact operator norm of the functional

$$a^\circ \mapsto \mathcal{R}_H(a, b)$$

on the centered subspace of $\ell^2(G)$.

Proof. Cauchy–Schwarz in (2) gives

$$\begin{aligned} |\mathcal{R}_H(a, b)| &\leq \frac{1}{n} \left(\sum_{\chi \neq \chi_0} |\widehat{a}(\overline{\chi})|^2 \right)^{1/2} \left(\sum_{\chi \neq \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 |\widehat{b}(\overline{\chi})|^2 \right)^{1/2} \\ &= \|a^\circ\|_2 \mathfrak{S}_H(b) \end{aligned}$$

by Parseval. Equality in Cauchy–Schwarz is attained by prescribing the nonprincipal Fourier coefficients of a proportional to the complex conjugates of

$$\widehat{\mathbf{1}_{E_H}}(\chi) \widehat{b}(\overline{\chi})$$

with the corresponding index reversal. Fourier inversion produces a centered a , proving exactness. $\square$

Theorem 2.3 (Wiener-to-Hilbert–Schmidt transfer). For every $b : G \rightarrow \mathbb{C}$,

$$\mathfrak{S}_H(b) \leq \|b\|_1 \sqrt{\frac{2H(q-1-2H)}{q-1}}, \quad (5)$$

$$|\mathcal{R}_H(a, b)| \leq \|b\|_1 \sqrt{\frac{2H(q-1-2H)}{q-1}} \|a^\circ\|_2. \quad (6)$$

Proof. The elementary Wiener bound gives

$$|\widehat{b}(\chi)| \leq \sum_{r \in G} |b(r)| = \|b\|_1.$$

TPC-99's exact nonprincipal Hilbert–Schmidt ledger is

$$\sum_{\chi \neq \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 = 2H(q-1-2H).$$

Insert both facts into (3), then use [theorem 2.2](#). □

Remark 2.4 (No character-sum theorem is used). The factor in (6) comes from the complete nonprincipal squared spectrum, not from Pólya–Vinogradov or a pointwise estimate for one character. It is therefore sensitive to the resonance width and to the purely principal endpoint.

Remark 2.5 (Why the asymmetric norm is legitimate). The theorem does not claim a smaller operator norm for arbitrary pairs (a, b) . It fixes b , uses its actual Wiener mass, and takes the norm only in the unknown census a . This is precisely the asymmetry of the literal reconstruction filter.

3 Sharpness and endpoint geometry

Put

$$g_q(H) = \sqrt{\frac{2H(q-1-2H)}{q-1}}. \tag{7}$$

Proposition 3.1 (Sharpness from a frequency atom). *Fix $r_0 \in G$, and let*

$$b = B\mathbf{1}_{\{r_0\}}, \quad B \in \mathbb{C}.$$

Then

$$\mathfrak{S}_H(b) = |B|g_q(H) = \|b\|_1 g_q(H).$$

Consequently (5) cannot be improved uniformly using only q, H , and $\|b\|_1$.

Proof. For every multiplicative character,

$$\widehat{b}(\chi) = B\overline{\chi(r_0)},$$

whose modulus is $|B|$. The assertion follows from (3) and the exact Hilbert–Schmidt ledger. □

Proposition 3.2 (The full-window endpoint is purely principal). *If $H = (q-1)/2$, then $E_H = G$,*

$$g_q(H) = 0, \quad \mathcal{R}_H(a, b) = 0$$

for every a, b . Thus

$$\mathcal{I}_H(a, b) = \left(\sum_{\omega} a(\omega) \right) \left(\sum_r b(r) \right).$$

Proof. The indicator of E_H is the constant function on G , so all of its nonprincipal Fourier coefficients vanish. The formula also follows directly from the fact that every pair (r, ω) is an edge. □

Remark 3.3 (Two-point fibers). For odd q ,

$$H_q(1) = H_q(2) = \frac{q-1}{2}.$$

Hence resolved fibers of length one or two have a full resonance window. Their multiplier distribution affects only the principal mass, not the centered spectral remainder. This exact endpoint was hidden by a uniform $\sqrt{q} \log q$ majorant.

Proposition 3.4 (Comparison with the worst-eigenvalue route). *For every H ,*

$$g_q(H) \leq \sqrt{2H} \leq \sqrt{q}.$$

If $H = o(q)$, the ratio between $g_q(H)$ and a $\sqrt{q} \log q$ cost tends to zero polynomially whenever $H \leq q^{1-\delta}$, apart from logarithms. At $H = (q-1)/2$, the ratio is exactly zero.

Proof. Drop the factor $(q-1-2H)/(q-1) \leq 1$ in (7); the second inequality uses $2H \leq q-1$. $\square$

Remark 3.5 (What sharpness means). **Proposition 3.1** is a sharp statement for the information $\|b\|_1$ alone. The actual reconstruction filter may have additional structure, so a still better filter-specific theorem is not ruled out. Conversely, one cannot simply delete $g_q(H)$ without using such new structure.

4 Return to the frequency-free literal census

We use the corrected TPC-99 crosswalk. Its base key is

$$\beta = (\theta, c, \kappa),$$

not the resolved key (θ, c, κ, r) containing a frequency. For each soluble nonconstant branch, put

$$w_{\beta,X} = |A_{\beta,X}| \mathcal{A}_{\beta,X},$$

where $A_{\beta,X}$ is the exact outer coefficient and $\mathcal{A}_{\beta,X}$ is the uniform absolute mass of the resolved z -fiber. Thus $w_{\beta,X}$ is independent of r and includes one copy of every source, content, projector, mask, and global-normalization factor.

For

$$1 \leq N_\beta \leq q_X, \quad H_\beta = H_{q_X}(N_\beta), \quad \omega_\beta = \varepsilon_\theta \frac{\Omega_\beta}{c} \pmod{q_X},$$

TPC-99 defines

$$a_H(\omega) = \sum_{\substack{\beta: H_\beta = H \\ \omega_\beta = \omega}} w_{\beta,X}. \quad (8)$$

The branch length N_β is the resolved z -fiber cardinality; j is already retained in θ . Branches with $N_\beta > q_X$ have no resonant nonzero frequency.

The fixed positive frequency weight is

$$b_X(r) = |\mathbf{m}_{K,X}(r)| \mathbf{1}_{\{0 < d_{q_X}(r,0) \leq R_X\}}, \quad B_X = \|b_X\|_1 \ll_K 1.$$

The phase of $\mathbf{m}_{K,X}$ is lost only because this is the positive carrier, not the original signed low window.

Theorem 4.1 (Width-sensitive literal carrier bound). *Let $\mathcal{C}_{\text{res},X}^{\text{abs}}$ be the corrected TPC-99 positive nonconstant resonance carrier. Then*

$$\begin{aligned} \mathcal{C}_{\text{res},X}^{\text{abs}} &\leq B_X \sum_H \frac{2H}{q_X - 1} \|a_H\|_1 \\ &\quad + B_X \sum_H g_{q_X}(H) \|a_H^\circ\|_2. \end{aligned} \quad (9)$$

In particular, it is $O(X^{o(1)} Q_X^2)$ if

$$\sum_H \frac{H}{q_X} \|a_H\|_1 \ll X^{o(1)} Q_X^2, \quad (10)$$

$$\sum_H g_{q_X}(H) \|a_H^\circ\|_2 \ll X^{o(1)} Q_X^2. \quad (11)$$

Proof. TPC-99 gives the exact positive-census identity

$$\mathcal{C}_{\text{res},X}^{\text{abs}} = \sum_H \mathcal{I}_H(a_H, b_X).$$

Apply (1) and then [theorem 2.3](#) to every width. The first gate is sufficient for the displayed principal term because $q_X - 1 \asymp q_X$ and $B_X \ll_K 1$; the second controls the remaining sum. $\square$

Corollary 4.2 (Strict weakening of the old centered gate). *The TPC-99 condition*

$$\sum_H \|a_H^\circ\|_2 \ll \frac{X^{o(1)} Q_X^2}{\sqrt{q_X} \log q_X}$$

implies (11). The converse is false even at the level of abstract nonnegative censuses.

Proof. Use $g_{q_X}(H) \leq \sqrt{q_X}$; the logarithm is a soft factor. For the converse, take a nonnegative point-mass census at one width $H = O(1)$, with centered norm of order Q_X^2 . The new left side is $O(Q_X^2)$, whereas the old target requires the norm to be $O(Q_X^{3/2+o(1)})$. $\square$

Remark 4.3 (No positive carrier identity with the signed sum). The crosswalk covers every frequency-free branch once and preserves its exact magnitude and support. It still uses $|A_{\beta,X} S_{\beta,X}(r) \mathbf{m}_{K,X}(r)|$, so (9) is an upper bound, not an equality with the signed affine Möbius contribution.

5 Resolved-length interpretation

Recall

$$H_q(N) = \min \left\{ \frac{q-1}{2}, \left\lfloor \frac{q}{N} \right\rfloor \right\}.$$

Proposition 5.1 (Exact endpoint and long-fiber gain). *For $1 \leq N \leq q$,*

$$g_q(H_q(N)) = 0, \quad N = 1, 2, \quad (12)$$

$$g_q(H_q(N)) \leq \sqrt{\frac{2q}{N}}, \quad N \geq 3. \quad (13)$$

Proof. For $N = 1, 2$, one has $H_q(N) = (q-1)/2$, so [proposition 3.2](#) applies. For $N \geq 3$,

$$H_q(N) \leq \frac{q}{N}.$$

Combine this with $g_q(H) \leq \sqrt{2H}$. $\square$

To expose the gain without merging lengths, refine (8) by the exact resolved cardinality:

$$a_N(\omega) = \sum_{\substack{\beta \text{ in the corrected nonconstant carrier} \\ 1 \leq N_\beta = N \leq q_X \\ \omega_\beta = \omega}} w_{\beta,X}. \quad (14)$$

Then

$$a_H = \sum_{\substack{1 \leq N \leq q_X \\ H_{q_X}(N) = H}} a_N, \quad a_H^\circ = \sum_{\substack{1 \leq N \leq q_X \\ H_{q_X}(N) = H}} a_N^\circ.$$

Corollary 5.2 (A resolved-length sufficient gate). *The centered condition (11) follows from*

$$\sqrt{q_X} \sum_{N=3}^{q_X} \frac{\|a_N^\circ\|_2}{\sqrt{N}} \ll X^{o(1)} Q_X^2. \quad (15)$$

The exact-length contributions $N = 1, 2$ incur no centered cost.

Proof. Use the triangle inequality in the preceding decomposition and then (13). The harmless factor $\sqrt{2}$ is absorbed in the implied constant. $\square$

Remark 5.3 (This is a sufficient refinement). Splitting a_H into exact lengths can lose cancellation between different N having the same width. Therefore (15) is a transparent sufficient condition, not a necessary reformulation of (11).

Remark 5.4 (Polynomial meaning). If a packet is supported on resolved lengths $N \geq N_{\min} \geq 3$, its centered spectral factor is at most

$$O\left(\sqrt{\frac{q_X}{N_{\min}}}\right).$$

Thus a polynomial lower bound for genuine fiber length translates directly into a polynomial saving over the previous uniform $\sqrt{q_X}$ cost. Establishing such support for the full literal packet is a separate physical statement.

6 Cell energy and a principal-to-diagonal transfer

TPC-100 first returns the $q_X \mid u$ subcarrier at $O(X^{o(1)}Q_X^2)$. It then refines the remaining $q_X \nmid u$ mass into opened atoms and groups those atoms into lossless cells while retaining the parent branch length N_β . Write a_H^{inv} for this invertible-ultra census. For every exact width H , TPC-100 proves

$$\|(a_H^{\text{inv}})^\circ\|_2^2 = \mathfrak{D}_H^\circ + \mathfrak{X}_H^\circ, \quad \mathfrak{D}_H^\circ \geq 0, \quad (16)$$

where $\mathfrak{D}_H^\circ$ is the one-cell diagonal ledger and $\mathfrak{X}_H^\circ$ is the centered excess from collisions between distinct cells.

At the maximally literal TPC-100 resolution, a cell retains (m, u, σ) , so the source identity recovers

$$j = \frac{\sigma u - h_0}{m}.$$

Every such fine cell is therefore a singleton. Consequently

$$\mathfrak{D}_H^\circ = \frac{q_X - 2}{q_X - 1} \sum_{\substack{p=(\beta, z): H_\beta=H \\ q_X \nmid u_p}} w_{\beta, z}^2.$$

This is an exact atomic diagonal ledger, not an intra-cell dispersion gain; every collision between distinct opened atoms remains in $\mathfrak{X}_H^\circ$.

Theorem 6.1 (Width-weighted cell gates). *After the $q_X \mid u$ return, the full principal condition (10) (or its restriction to the invertible census), together with the two estimates*

$$\sum_H g_{q_X}(H) \sqrt{\mathfrak{D}_H^\circ} \ll X^{o(1)} Q_X^2, \quad (17)$$

$$\sum_H g_{q_X}(H) \sqrt{|\mathfrak{X}_H^\circ|} \ll X^{o(1)} Q_X^2 \quad (18)$$

are sufficient to return the remaining invertible positive nonconstant resonance carrier. Adding the already returned $q_X \mid u$ piece gives $O(X^{o(1)}Q_X^2)$ for the complete positive nonconstant carrier.

Proof. By (16),

$$\|(a_H^{\text{inv}})^\circ\|_2 \leq \sqrt{\mathfrak{D}_H^\circ} + \sqrt{|\mathfrak{X}_H^\circ|}.$$

Multiply by $g_{q_X}(H)$, sum over H , and apply [theorem 4.1](#). $\square$

There is an abstract way to discharge the diagonal condition from the principal ledger. It isolates exactly the additional local fact that must be checked in the literal weights.

Definition 6.2 (Opened-atom cap). Let $w_C(j) \geq 0$ be the normalized $q_X \nmid u$ opened-atom weights used in the TPC-100 cell census, so

$$M_H^{\text{inv}} = \|a_H^{\text{inv}}\|_1 = \sum_{C,j} w_C(j).$$

Put

$$W_X = \max_{H,C,j} w_C(j)$$

over the actual support, with the convention $W_X = 0$ when that support is empty, and let $\mathcal{H}_X = \{H : g_{q_X}(H) > 0, M_H^{\text{inv}} > 0\}$.

Theorem 6.3 (Principal-to-diagonal transfer). *Define the principal mass without the bounded filter factor by*

$$\mathfrak{P}_X^{\text{inv}} = \sum_H \frac{2H}{q_X - 1} M_H^{\text{inv}}.$$

Then

$$\boxed{\sum_H g_{q_X}(H) \sqrt{\mathfrak{D}_H^\circ} \leq \sqrt{W_X |\mathcal{H}_X| (q_X - 1) \mathfrak{P}_X^{\text{inv}}}.} \quad (19)$$

Consequently, if

$$q_X \asymp Q_X, \quad W_X \leq X^{o(1)}, \quad \mathfrak{P}_X^{\text{inv}} \leq X^{o(1)} Q_X^2,$$

then (17) follows automatically.

Proof. The diagonal definition and nonnegativity give

$$0 \leq \mathfrak{D}_H^\circ \leq \sum_{C,j} w_C(j)^2 \leq W_X M_H^{\text{inv}}.$$

Therefore Cauchy–Schwarz over the active widths yields

$$\begin{aligned} \sum_H g_{q_X}(H) \sqrt{\mathfrak{D}_H^\circ} &\leq \sqrt{W_X} \sum_{H \in \mathcal{H}_X} g_{q_X}(H) \sqrt{M_H^{\text{inv}}} \\ &\leq \sqrt{W_X |\mathcal{H}_X| \sum_H g_{q_X}(H)^2 M_H^{\text{inv}}}. \end{aligned}$$

Since

$$g_{q_X}(H)^2 \leq 2H,$$

the last sum is at most $(q_X - 1) \mathfrak{P}_X^{\text{inv}}$, proving (19). Finally $|\mathcal{H}_X| \leq (q_X - 1)/2$ and $q_X \asymp Q_X$. The full principal mass dominates $\mathfrak{P}_X^{\text{inv}}$ because the opened weights are nonnegative. $\square$

Remark 6.4 (A real reduction, not a hidden proof). Once the literal atom cap $W_X \leq X^{o(1)}$ and the principal gate are verified, no separate diagonal-energy theorem is needed. This paper does not assert that cap: an opened atom includes its actual row coefficients, exact-content and projector factors, smooth weights, and the single global normalization. Bounding that specific quantity is the next literal audit, not a formal consequence of row ℓ^1 mass.

Remark 6.5 (The cross-cell door remains). The transfer does not control $\mathfrak{X}_H^\circ$. TPC-100 shows that its support obeys an exact affine collision congruence, but distinct cells can still align maximally. Since the fine cells are singleton, this includes all collisions between distinct atoms, even two atoms under the same parent branch. The surviving positive route is therefore the weighted cross-cell gate (18), or a return to signed arithmetic before the positive census is formed.

7 Exact finite regression

The script `experiments/tpc101_certificate.py` enumerates odd primes, every admissible width, complete multiplicative-character tables, and deterministic complex test weights. It verifies:

1. the direct incidence remainder against (2);
2. the exact fixed-filter norm inequality and an explicitly constructed Fourier extremizer;
3. the Wiener-to-Hilbert–Schmidt bound;
4. equality for a one-frequency atom;
5. the full-window zero remainder and the resolved-length bound; and
6. the principal-to-diagonal transfer on deterministic finite opened-cell ledgers.

Roots of unity are evaluated in floating arithmetic; all set, cardinality, endpoint, and ledger data are deterministic. The reported roundoff tolerance is numerical verification of finite identities, not evidence for an asymptotic census bound.

8 Route and exponent ledger

The positive exceptional route now reads

piece	status
conductor one	$O(X^{o(1)}Q_X^2)$ in TPC-98
$q_X \nmid \Omega, q_X \mid u$	$O(X^{o(1)}Q_X^2)$ in TPC-100
principal nonconstant resonance	$\sum_H \frac{2H}{q_X - 1} M_H^{\text{inv}}$ gate
centered nonconstant resonance	$\sum_H g_{q_X}(H) \ (a_H^{\text{inv}})^\circ\ _2$ gate.

The centered row is strictly sharper than the TPC-99 worst-eigenvalue return. Under a subpower opened-atom cap, its diagonal part follows from the principal row by [theorem 6.3](#). The remaining positive obligation is therefore:

1. verify the literal principal mass;
2. verify the normalized opened-atom cap; and
3. control the width-weighted cross-cell excess (18).

If the atom cap fails at polynomial scale, it must be entered in the physical budget and the diagonal gate reopens. If proof-carrying cross-cell alignment makes (18) fail at a polynomial scale, this termwise positive modulus-incidence implementation stops and one must use a sharper filter-sensitive bound or retain signed affine arithmetic.

The ledgers remain independent:

$$\lambda_D \leq 2\eta_Z$$

is the determinant/zero-mode compatibility, whereas

$$\Lambda_{\text{phys}} < \frac{1}{400}$$

is the strict endpoint-loss condition. No part of the Hilbert–Schmidt gain is charged twice as a physical exponent, and equality at $1/400$ remains a stop.

TPC-18 full-block reassembly and primitive determinant dispersion remain independent. The present theorem improves one resonance operator estimate; it proves neither global reassembly nor fixed- h_0 Möbius cancellation.

9 Claim boundary

Proved

- The exact fixed-filter functional norm $\mathfrak{S}_H(b)$ is unconditional finite-group algebra.
- Bounded Wiener mass transfers the complete nonprincipal Hilbert–Schmidt ledger to the sharp factor $g_q(H)$.
- A frequency atom proves sharpness using only $\|b\|_1$.
- Full resonance width is purely principal, and resolved length $N \geq 3$ gives the factor $O(\sqrt{q/N})$.
- The corrected frequency-free TPC-99 positive carrier obeys the width-sensitive principal/centered return (9).
- After the separate $q_X \mid u$ return, the explicitly stated opened-atom cap and principal gate imply the width-weighted diagonal gate for the remaining invertible census. The fine cells are singleton, so this is an atomic-cap transfer and not a new intra-cell dispersion theorem.

The finite-group statements are L0. The proof-carrying literal census crosswalk and the conditional atom-ledger reduction are L1.

Not proved

- The growing literal principal gate (10) is not proved.
- The maximum normalized opened-atom bound $W_X \leq X^{o(1)}$ is isolated, not asserted.
- The width-weighted cross-cell gate (18) is not proved.
- No positive carrier is identified with the original signed low window.
- No new affine Möbius cancellation theorem, determinant lower bound, TPC-18 cover, parity breakthrough, fixed-shift Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime conclusion is proved or implied.
- Nothing is specialized to $h_0 = 2$.

The paper therefore removes a genuine worst-eigenvalue loss and narrows the next literal obligations, but it does not claim L2 arithmetic progress.

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